

YEAR 7 · UNITS 2 AND 3 · HOMEWORK 8

Where It Lands on the Line

Science 2.8 and 2.7 – the pH scale; compounds and mixtures

Mathematics 2.6, 2.5 and 3.1–3.2 – inequalities, equations, rounding

Section F – the pH scale, the cylinder and the graph are one thing

Name: _____

Class: _____

Date given: _____

Date due: **Monday 28 September 2026****How to do this homework**

Six sections. Science first this time: the maths in Section C reads the science table in Section A.

- In science, write **full English sentences**. “Acid” is not a sentence; “Lemon juice is an acid because its pH is below 7” is.
- In maths, **write the equation before you write the answer**. **No calculator**, except where Section E says so.

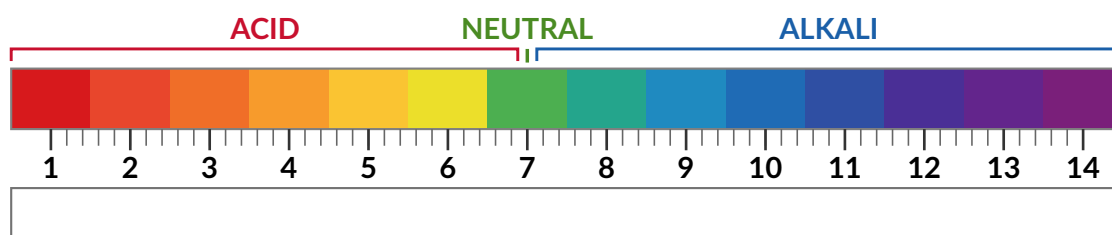
Section A · Science 2.8 – Acids, Bases and the pH Scale

A1 Put each liquid on the scale

Below is the pH scale, coloured the way universal indicator colours it.

Word help – five words the whole section runs on

acid	a substance that tastes sour. A strong one burns skin
base · alkali	the chemical opposite of an acid. An alkali is a base dissolved in water , and it is the word the exam uses
neutral	not an acid and not an alkali. pH exactly 7
indicator	a substance that changes colour to show an acid or an alkali



(a) Every pH below is a **decimal**. Draw a line on the scale where it goes and write the letter

in the white strip – there are **five small spaces** between each pair of numbers. **(b)** Finish the last column.

	Liquid	pH	Acid, neutral or alkali?		Liquid	pH	Acid, neutral or alkali?
A	lemon juice	2.4		E	sea water	8.2	
B	orange juice	3.8		F	baking soda	9.6	
C	black coffee	5.2		G	soap	10.8	
D	pure water	7.0		H	bleach	12.6	

A2 Why the school buys both

Litmus is cheap. Universal indicator costs more. Mr Bowen buys both.

Two colours, or a number

Litmus has **two** colours only: red in an acid, blue in an alkali. **Universal indicator** gives a **different colour for every pH number**, so it gives you a number instead of just a word.

Mr Bowen tests two liquids with **red litmus paper**. Both turn it blue. One is pH 8 and the other is pH 13. Explain why litmus **cannot** tell him which is which, and say which indicator he should use instead.

Start: "**Litmus can only show...**" then "**...because both liquids are...**"

A3 The end that nobody worries about

A student says: "*Acids are the dangerous ones. pH 13 is safe, because 13 is a big number.*"

Mr Bowen keeps a bottle of oven cleaner under his kitchen sink. Its pH is **13**. Explain why the student is wrong. Use the word **corrosive**.

A4 Neutralisation

What neutralisation does

Add a **base** to an **acid** and they cancel each other out. The acid **stops being an acid** – the pH moves **towards 7**.

(a) Mr Bowen's garden soil has a pH of 4, and rice will not grow in it. He must move the pH **up** to about 7. Tick the **one** thing he should dig into the soil.

What he could add	Tick one	What he could add	Tick one
vinegar, pH 3		lime powder, pH 12	
lemon juice, pH 2		pure water, pH 7	

(b) Explain your choice in full sentences. Say what happens to the pH of the soil.

Section B · Science 2.7 – Compounds and Mixtures

B1 Compound or mixture?

Tick one box in each row.

The test is whether you can get it back

A **mixture** is stirred together, so you can **separate** it again – with a magnet, a filter, or by boiling. A **compound** is **bonded** together: it is a **new substance**, and a magnet will not undo it.

	Compound	Mixture		Compound	Mixture
air			water, H ₂ O		
salt, NaCl			sea water		
sand stirred into salt			carbon dioxide, CO ₂		
mineral water			iron sulfide, FeS		

B2 Iron and sulfur, twice

Mr Bowen puts iron filings and yellow sulfur powder in a dish and **stirs** them. Then he puts a second lot in a test tube and **heats** it until it glows.

He holds a magnet over each one. The magnet pulls the iron out of the **first** dish. It does nothing at all to the **second**.

Explain, in full sentences, what heating did and why the magnet stopped working. Start: "**Be-fore it was heated, the iron and the sulfur were...**" then "**Heating made the atoms...**"

Section C · Mathematics 2.6 – Inequalities

C1 Say it in English

Write $<$ or $>$ in each box, then write the whole thing as an English sentence. The first one is done for you.

Word help – the words a question uses instead of the signs

is less than $\cdot <$ also fewer than, below, under. *Under 18* means age < 18

is greater than $\cdot >$ also more than, above, over. **Greater** is not **great**

The numbers	$<$ or $>$	Say it in English
6 11	$<$	6 is less than 11
9 2		
-4 1		
-7 -3		

C2 Which liquids does the inequality mean?

Here are the eight liquids from A1 again, with their pH.

A	B	C	D	E	F	G	H
lemon juice	orange juice	black coffee	pure water	sea water	baking soda	soap	bleach
2.4	3.8	5.2	7.0	8.2	9.6	10.8	12.6

The letter p stands for the pH of a liquid. For each inequality, write the letters of **every** liquid that works.

The number in the inequality never works itself

$p > 9.6$ means **greater than** 9.6. A liquid at **exactly** 9.6 does **not** count. This is the open circle: the circle is empty because that number is left out.

One of the six has **no answer at all**. If you find it, write **none**.

(a) $p < 7$

(b) $p > 7$ _____

(c) $p > 9.6$ _____

(d) $p < 3.8$ _____

(e) $p > 5$ and $p < 9$ _____

(f) $p > 12.6$ _____

C3 Now write the inequality

Each sentence is a real rule. Write it using $<$ or $>$. Use the letter given.

(a) A swimming pool is safe when its pH is **above** 7.2. Use p . _____

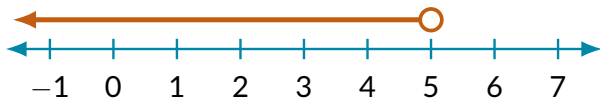
(b) Rain is called acid rain when its pH is **below** 5.6. Use r . _____

(c) Rice grows when the soil pH is **more than** 5. Use s . _____

(d) A lab liquid is corrosive when its pH is **under 2 or over 12**. Use c . Write **two** inequalities. _____

C4 Read the line, then draw the line

(a) Write the inequality each number line shows. Use x .

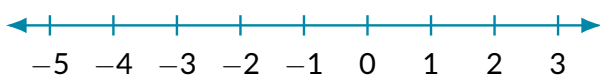


(i) _____



(ii) _____

(b) Now draw this one. Open circle, then an arrow.



$$m > -2$$

C5 The smallest one that works

An **integer** is a whole number: negative, zero or positive.

(a) $x > 6$. Smallest integer? _____

(c) $n > -1$. Smallest integer? _____

(b) $y < -4$. Largest integer? _____

(d) $t > 2.5$. Smallest integer? _____

Section D · Mathematics 2.5 — Equations and Formulae

D1 Put it back in and see

Someone has already written an answer. Put it back into the equation. Tick **right** or **wrong**, and if it is wrong, write the correct answer.

Checking is one line of work

To check $x - 7 = 5$ when somebody says $x = 2$, **put 2 in:** $2 - 7 = -5$, not 5. So it is wrong.

Equation	Their answer	Right	Wrong	If wrong, write it
$x - 7 = 5$	$x = 2$			
$3y = 24$	$y = 8$			
$m + 9 = 15$	$m = 24$			
$k - 6 = 10$	$k = 4$			

D2 Two steps

Undo the **last** step first. Write the middle line.

Last on, first off

$2a + 4 = 18$. The last thing done to a was **add 4**, so undo that first.

$18 - 4 = 14$, then $14 \div 2 = 7$. Check: $2 \times 7 + 4 = 18$. **Not $18 \div 2$ first.**

- | | | | |
|--------------------|-------------|-------------------|-------------|
| (a) $3a + 5 = 26$ | $a =$ _____ | (d) $2d - 9 = 15$ | $d =$ _____ |
| (b) $4b - 7 = 25$ | $b =$ _____ | (e) $6m + 4 = 40$ | $m =$ _____ |
| (c) $5c + 12 = 47$ | $c =$ _____ | (f) $8n - 3 = 45$ | $n =$ _____ |

D3 You are given the formula. Find the letter.

In each one you are told the **answer** the formula gives. Work backwards to find the letter. **Write the equation first.**

Two jobs, in this order

A taxi costs $C = 15n + 20$ where n is the number of kilometres. Mr Bowen's taxi cost **95**. How far did he go?

1. Put the number in: $15n + 20 = 95$

2. Solve it: $95 - 20 = 75$, then $75 \div 15 = 5$. He went 5 km.

(a) $P = 4s$ (perimeter of a square, side s). $P = 52$. $s =$ _____

(b) $C = 8n + 15$ (cost of n tickets). $C = 95$. $n =$ _____

(c) $d = 60t$ (km travelled in t hours). $d = 480$. $t =$ _____

(d) $V = 30h$ (cm^3 in a tank at depth h cm). $V = 420$. $h =$ _____

(e) $m = 60 + v$ (grams: a beaker plus $v \text{ cm}^3$ of water). $m = 145$. $v =$ _____

(f) $F = 2n - 6$. $F = 38$. $n =$ _____

D4 Word problems

Write the equation, then solve it. Show your working in the box.

(a) Mr Bowen's jug holds 340 cm^3 of water. He pours the **same amount** into each of **6** beakers, and 40 cm^3 is left in the jug. How much water went into each beaker?

(b) Mr Bowen has some fish. He buys **4** more. Then his cat eats **half** of all of them. He now has **6** fish. How many did he have at the start?

Section E · Mathematics 3.1–3.2 – Place Value and Rounding**E1 Move the digits**

No calculator.

The point never moves – the digits do 7.2×10^3 is **7200**. **Not 7.2000**.Multiplying moves the digits **left**. Dividing moves them **right**. The power counts **how many places**.

(a) $3.7 \times 100 =$ _____ (c) $26 \times 10^4 =$ _____ (e) $63 \div 10^3 =$ _____

(b) $4.5 \times 10^3 =$ _____ (d) $940 \div 100 =$ _____ (f) $5 \div 10^4 =$ _____

E2 Round it**Keep the zero**

34.9892 to **1 decimal place** is **35.0**, not 35. The .0 is what shows you rounded to one place. Writing 35 loses the mark.

To 1 decimal place	7.34	12.87	0.952	29.96
To 2 decimal places	5.376	0.0961	8.2449	3.997

E3 One place further**Work out one extra digit, then round**

For $58 \div 7$ correct to **3 decimal places**, divide to **4**: 8.2857. The 4th digit is 7, so it is **8.286**, not 8.285. Calculator allowed here only.

- (a) $47 \div 9$, correct to 2 d.p. _____ (c) $26 \div 3$, correct to 3 d.p. _____
 (b) $45 \div 7$, correct to 2 d.p. _____ (d) $100 \div 7$, correct to 3 d.p. _____

E4 Mr Bowen's homework

One of these four is right. Find it, and correct the other three.

- (a) $6.4 \times 10^3 = 6.4000$ _____ (c) 2.996 to 2 d.p. = 2.99 _____
 (b) $380 \div 10^2 = 38$ _____ (d) 0.83 to 1 d.p. = 0.8 _____

Section F · Where it all meets — reading a scale

In **A1** you were given a number and had to find the place. Here it is the other way round: you are given the place and have to find the number. A cylinder is a number line stood on its end, and a graph has two — and both of them ask the same question first: **how much is one small line worth?**

F1 How much is one small line worth?

Count the spaces, not the lines

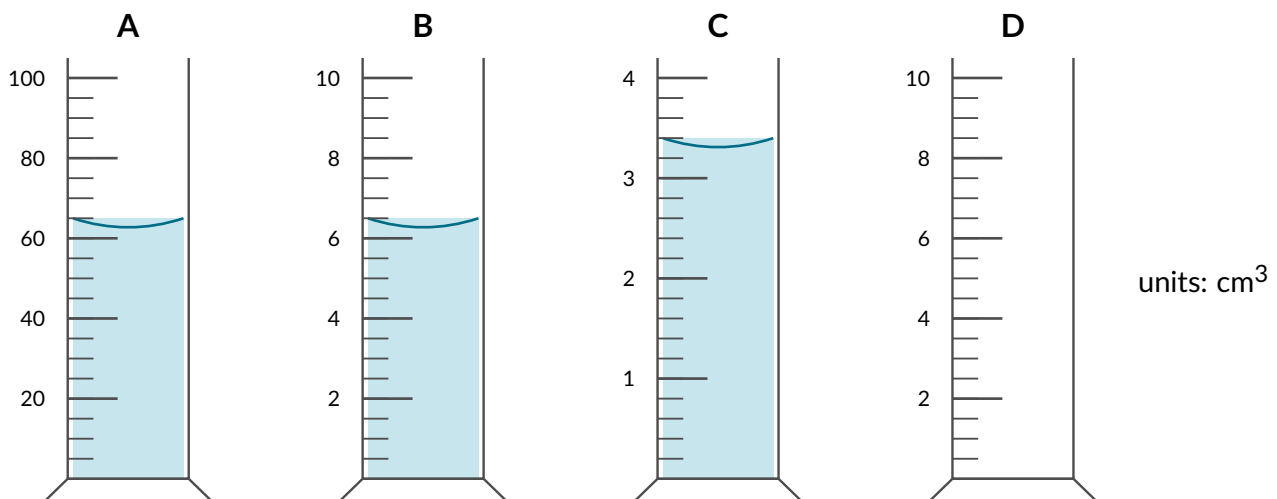
Look at **two numbered lines**. Count the **spaces** between them – the gaps, not the marks. Then **divide**. From **0 to 10** with **5 spaces**: $10 \div 5 = 2$, so one small line is **worth 2**.

A **scale** is the row of lines and numbers on an instrument. **Scales** is the machine you stand on to find your mass – a different word.

From	To	Spaces between them	One small line is worth
0	10	5	2
0	1	10	
20	30	4	
0	5	10	
50	100	5	

F2 Four measuring cylinders

Same small lines on all four; only the **numbers** differ. Read the bottom of the meniscus.

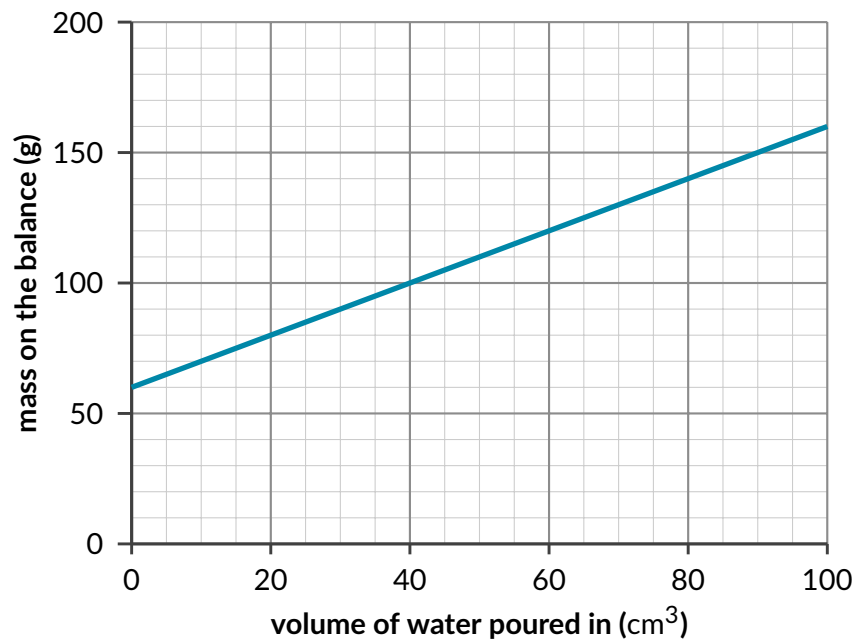


	One small line is worth	The volume of water is
A		
B		
C		
D		Draw the water at 8.5 cm³

(b) **A** and **B** are the same picture, and the water is at the same height. Explain why the two answers are not the same number.

F3 The same question, on a graph

Mr Bowen puts an **empty** beaker on a balance, then pours water in a little at a time and writes down the mass.



- (a) How much is one small line worth on the **across** axis? _____
- (b) How much is one small line worth on the **up** axis? _____
- (c) What is the mass when **40** cm³ has been poured in? _____
- (d) The balance reads **130** g. How much water is in the beaker? _____
- (e) What is the mass of the **empty** beaker? Say how you know.

- (f) Write a **formula** for the mass m grams when v cm³ of water has been poured in.

$m =$ _____